Math 133: Business Optimization

About 30 minutes

Name: _____ November 11, 2026

Use derivatives to justify each optimum on the stated interval, including relevant endpoints. Show the function you optimize and interpret your answer with units. Treat quantities as continuous. Keep exact values until the final step; a calculator may be used for approximations.

1. A computer company models its monthly profit by

$$P(x) = -x^2 + 24x - 44, \qquad 0 \le x \le 20,$$

where x is production in thousands of computers and $P(x)$ is profit in thousands of dollars. How many computers should it produce to maximize monthly profit, and what is that maximum profit?

2. A booth can supply at most 250 reusable mugs per day. Its price-demand model is

$$p(q) = 14 - 0.02q, \qquad 0 \le q \le 250,$$

where q is the number of mugs sold per day and $p(q)$ is the price in dollars per mug. Daily costs are \$100 plus \$2 per mug. What price should the booth charge to maximize daily profit? Give the number of mugs sold and the maximum daily profit as well.

3. A manufacturer models its weekly profit by

$$P(x) = -x^3 + 9x^2 - 15x - 10, \qquad 0 \le x \le 8,$$

where x is production in thousands of units and $P(x)$ is profit in thousands of dollars. Find and classify every critical production level in $(0, 8)$. Then determine how many units the manufacturer should produce to maximize weekly profit on $[0, 8]$, and state that maximum profit. Justify your choice over all feasible production levels.

4. A cosmetics company estimates weekly demand for a new skin-care product using the model

$$p(x) = 18e^{-x/3}, \qquad 0 \le x \le 9,$$

where x is the number of units sold per week in thousands and $p(x)$ is the price in dollars per unit. At what price should the company sell the product to maximize weekly revenue? Give the corresponding weekly sales quantity and maximum revenue. Give exact values, then round the price and revenue in dollars to the nearest cent.

5. A stadium concession stand estimates demand for hot dogs by

$$p(x) = 6 - \ln x, \qquad 1 \leq x \leq 60,$$

where x is the number of hot dogs sold per game in thousands and $p(x)$ is the price in dollars per hot dog. Each hot dog costs \$1, and fixed operating costs are \$5,000 per game. Determine the price that maximizes profit per game, the corresponding sales quantity, and the maximum profit. Give exact values and approximate the sales quantity to the nearest hot dog and the profit in dollars to the nearest cent.